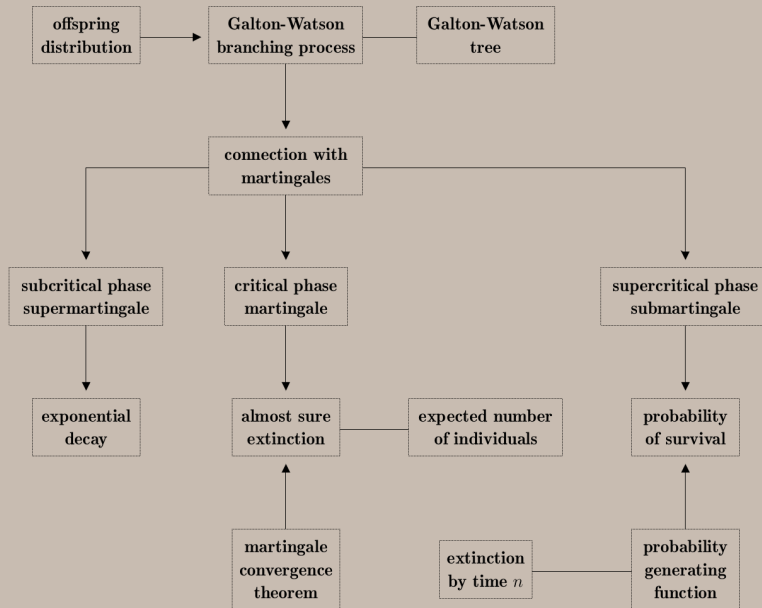
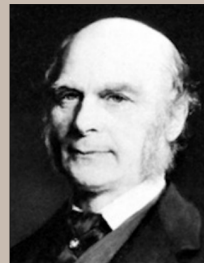
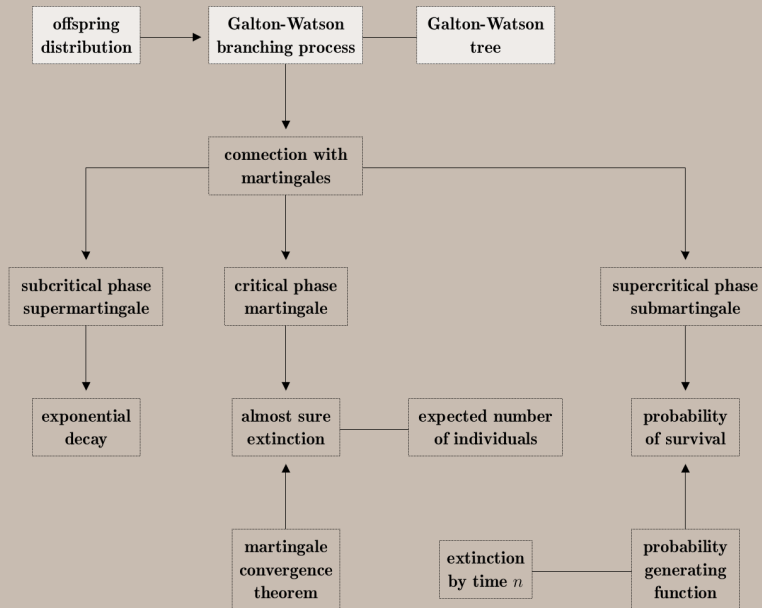


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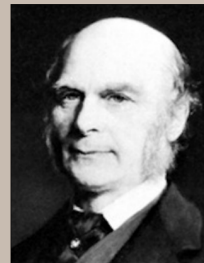
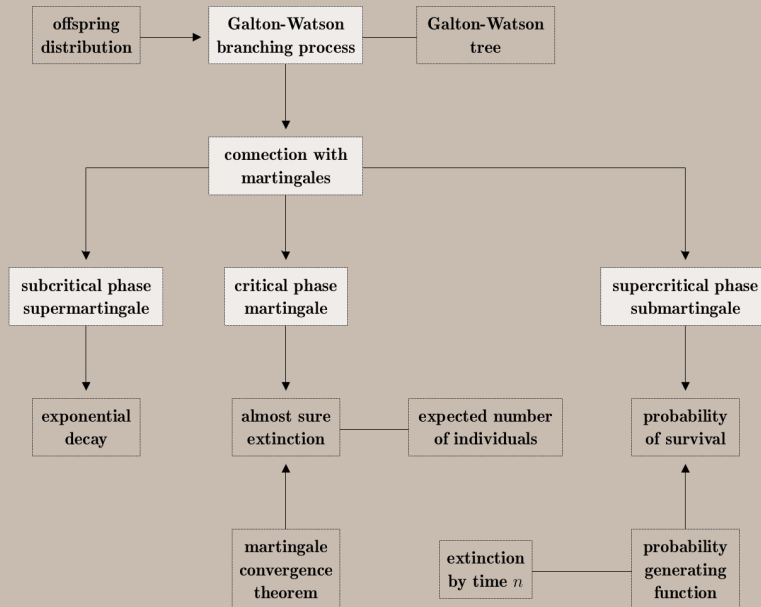
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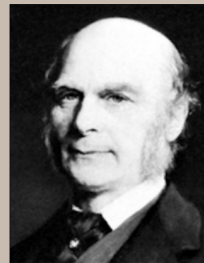
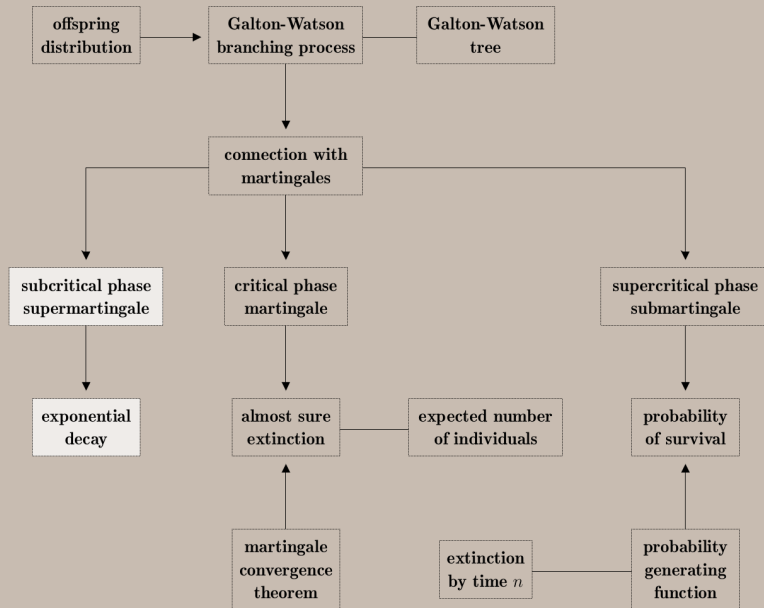
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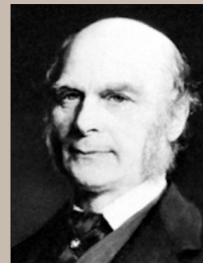
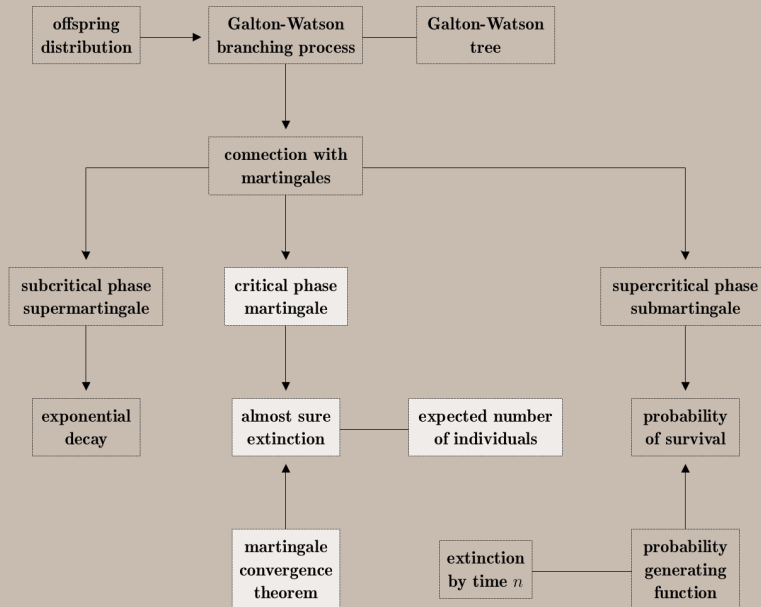
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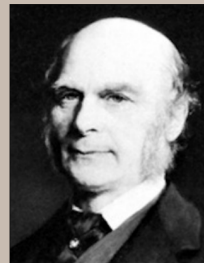
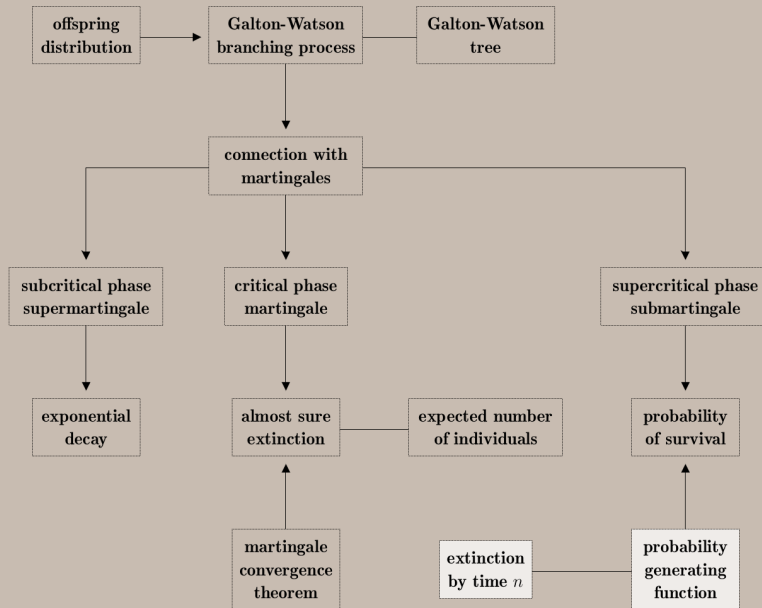
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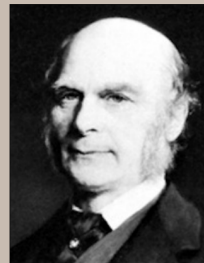
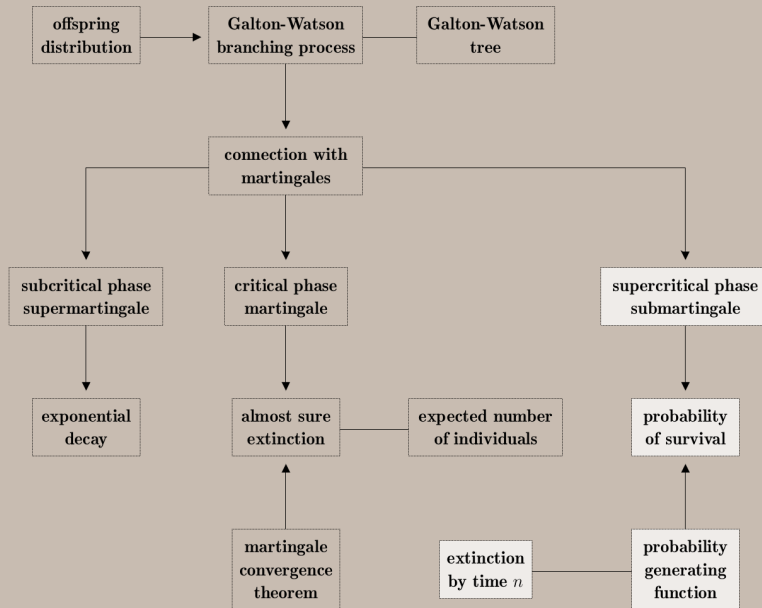
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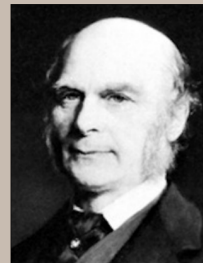
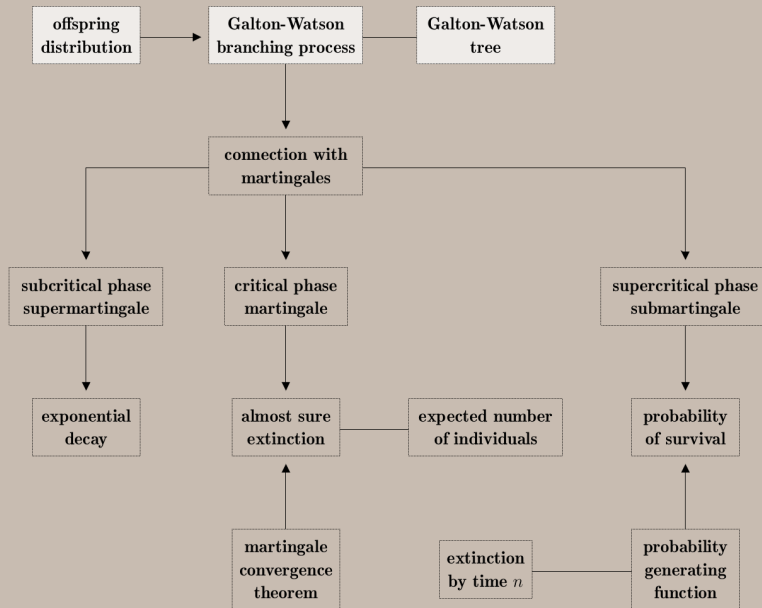
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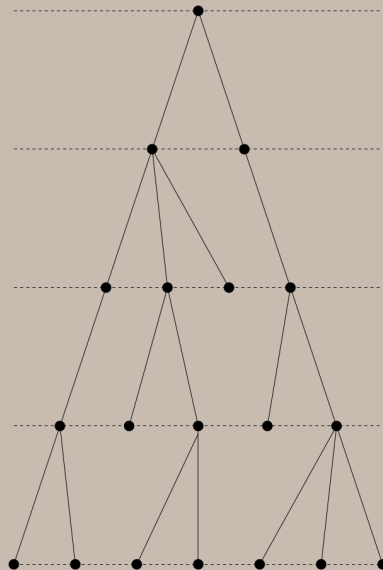
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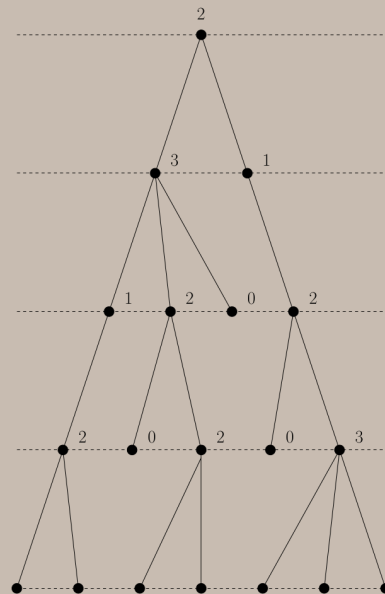
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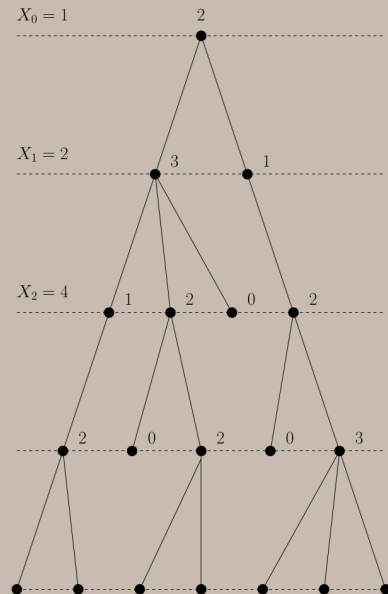
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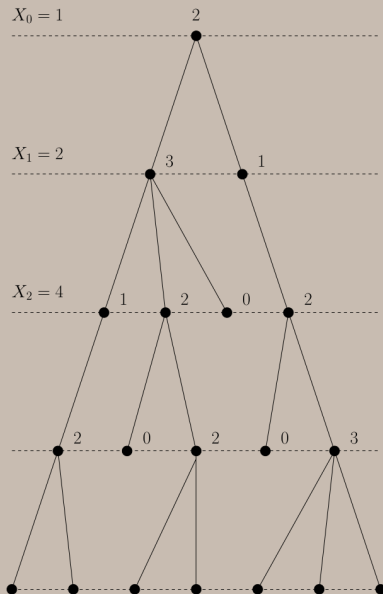


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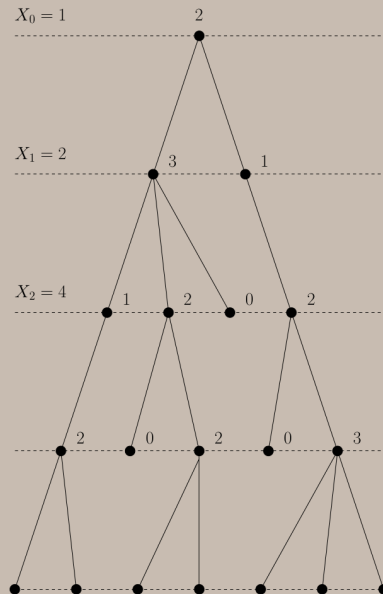
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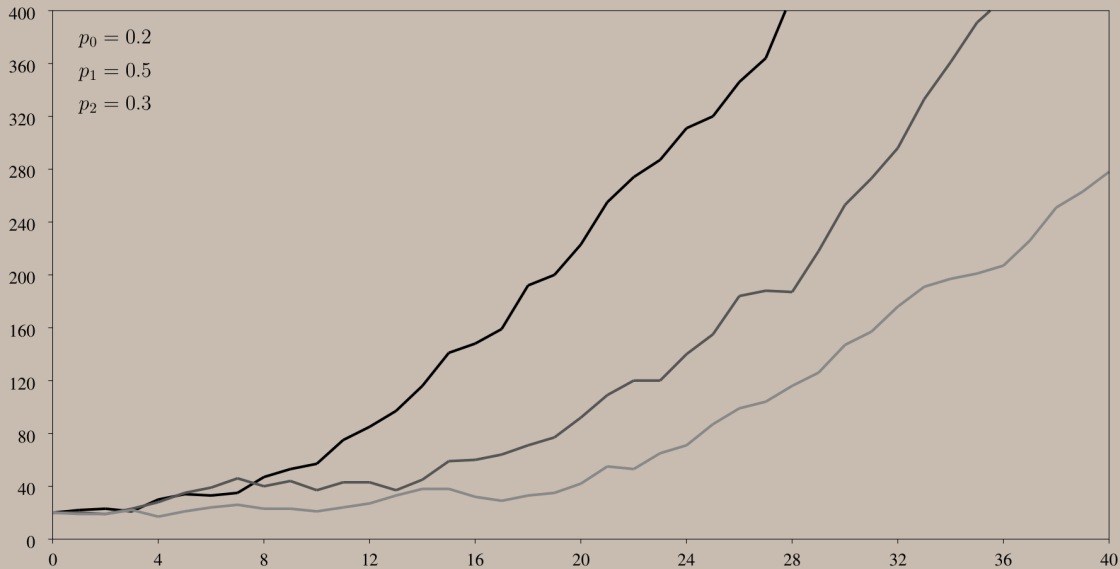
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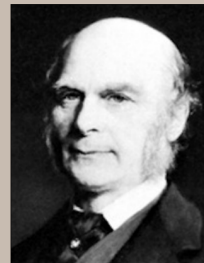
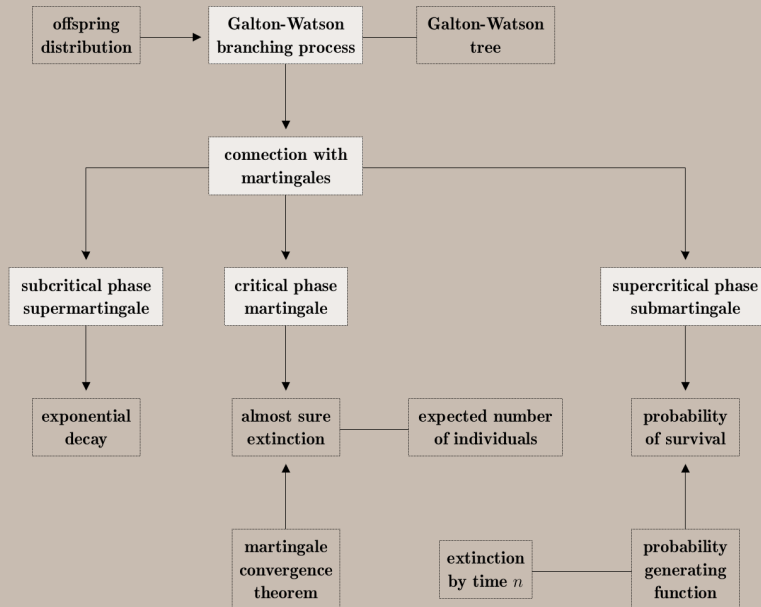
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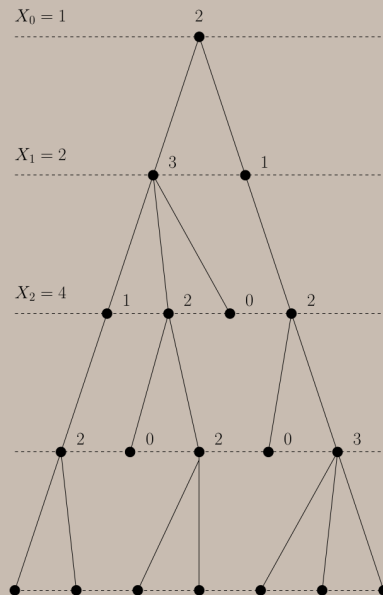
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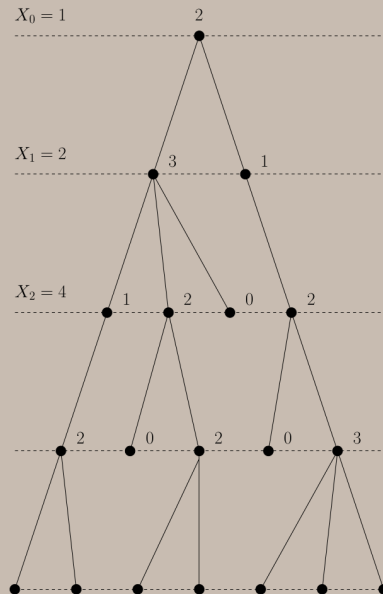
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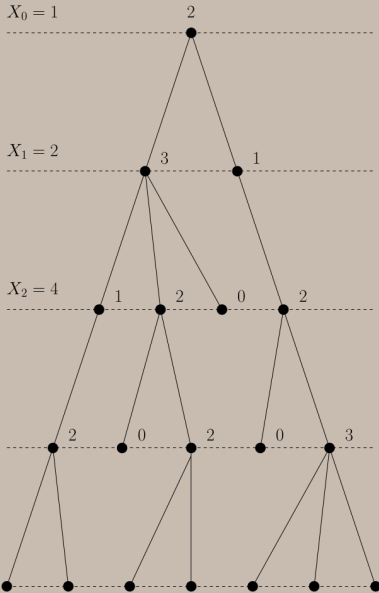
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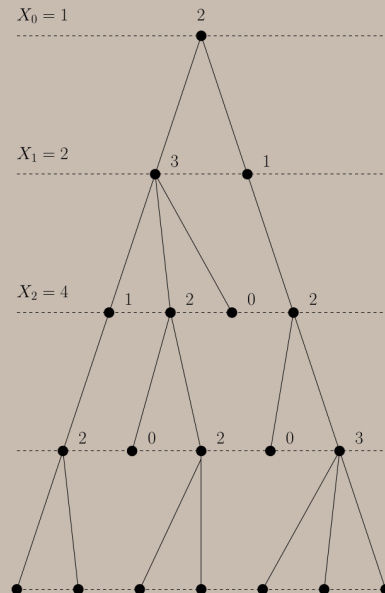
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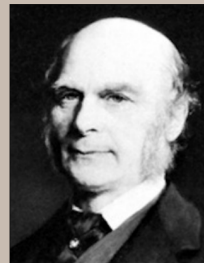
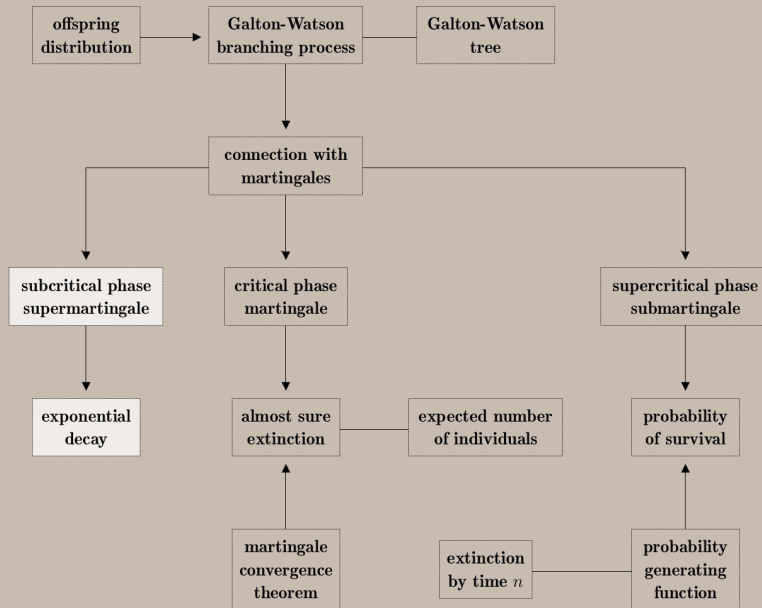
$\implies$ the rescaled process X_n/μ^n is a martingale.

subcritical phase $\mu < 1 \iff$ supermartingale,

critical phase $\mu = 1 \iff$ martingale,

supercritical phase $\mu > 1 \iff$ submartingale.





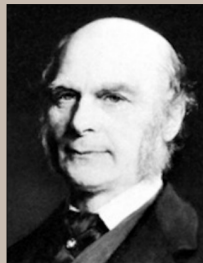
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Henry Watson

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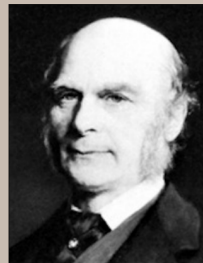


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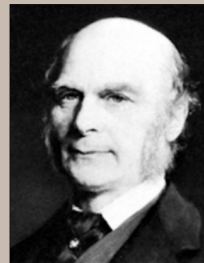
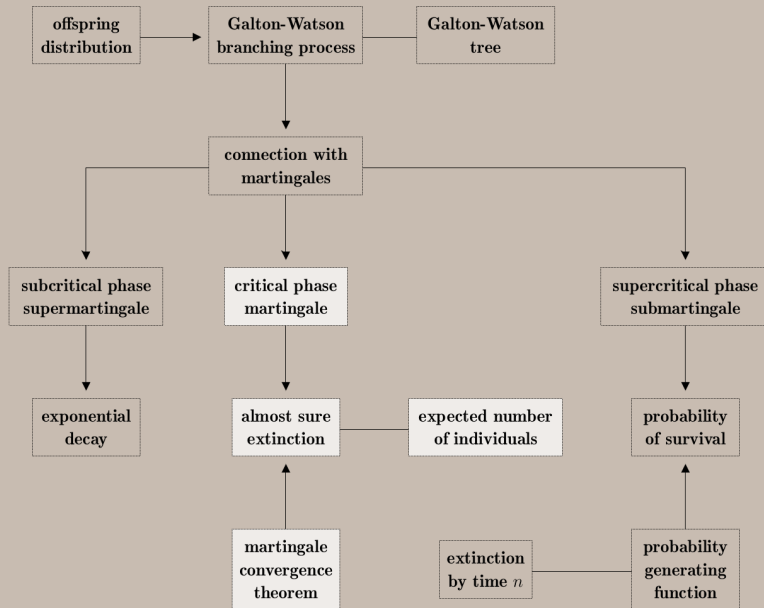
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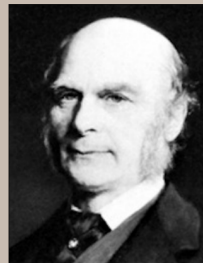
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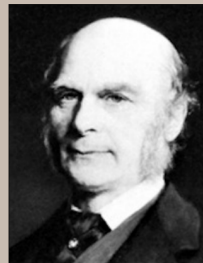
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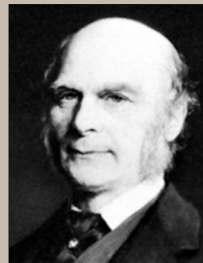
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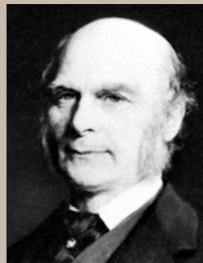
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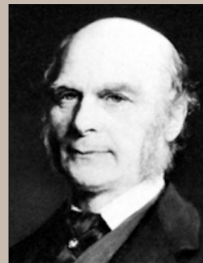
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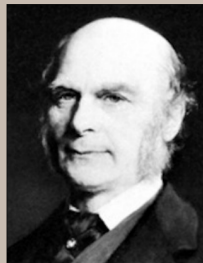
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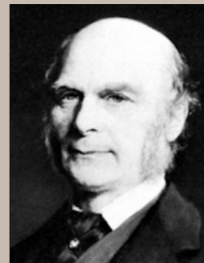
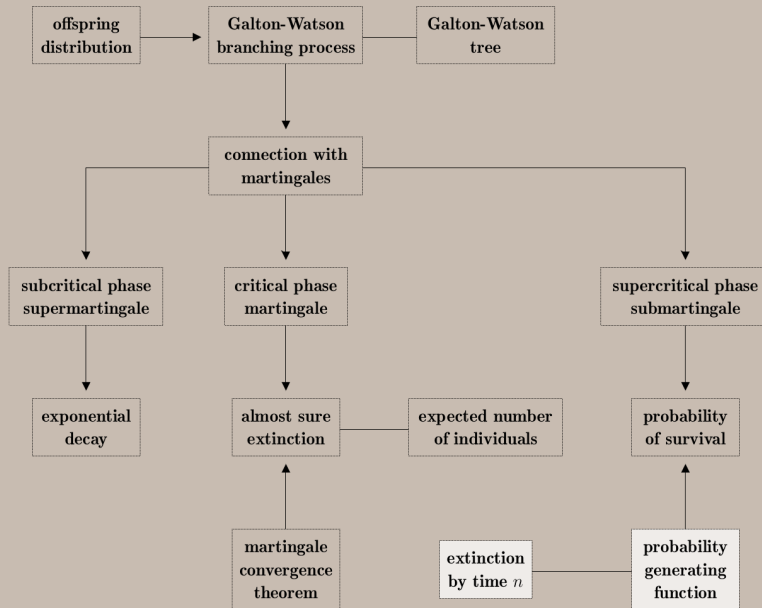
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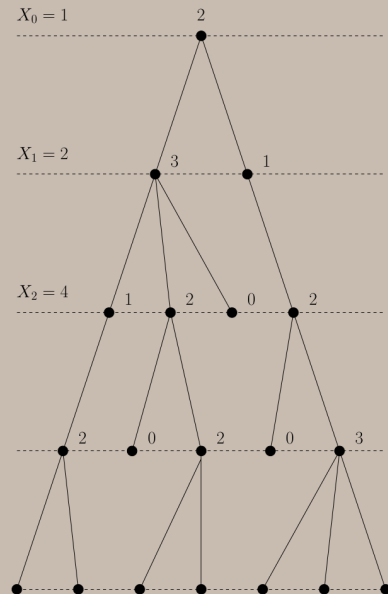
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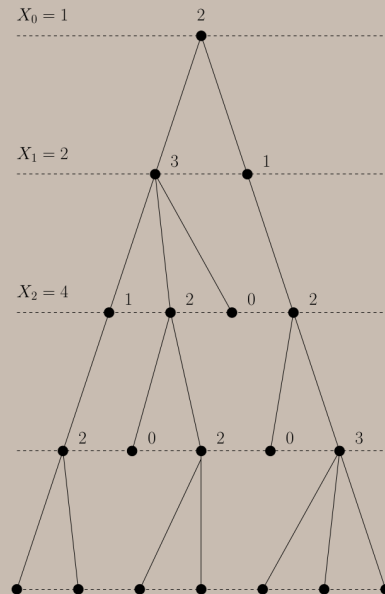
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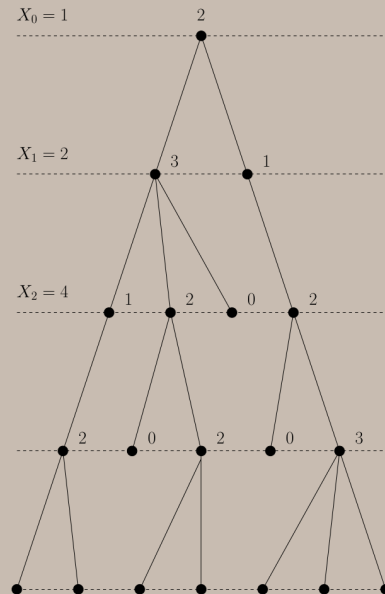
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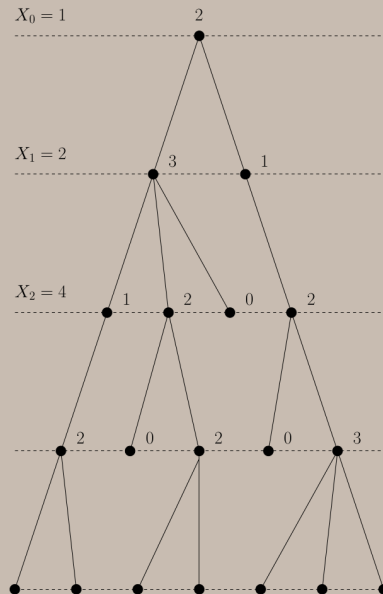
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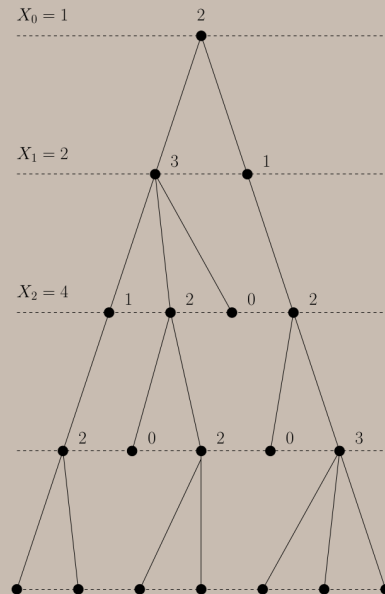
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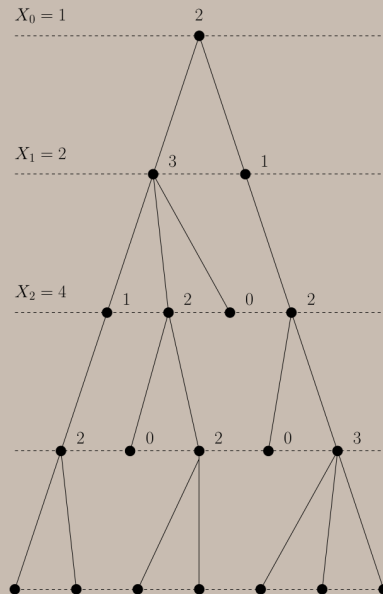


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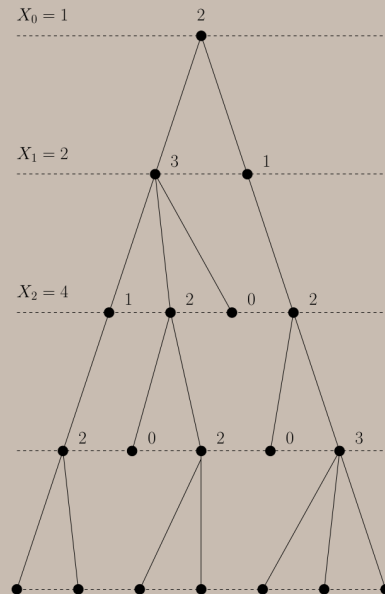
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$$\begin{aligned} \theta_{n+1} &= \sum_k P(X_{n+1} = 0 \mid X_1 = k) P(X_1 = k) \\ &\stackrel{\text{IND}}{=} \sum_k p_k [P(X_n = 0 \mid X_0 = 1)]^k \\ &= \sum_k p_k \theta_n^k = G(\theta_n). \end{aligned}$$

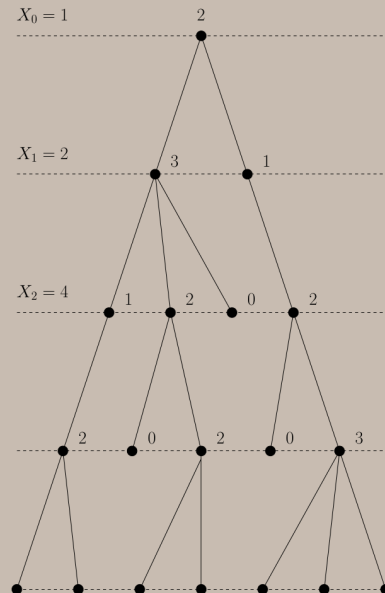
Monotonicity of the sequence $\theta_n \implies \theta_n \uparrow \theta_\infty$.

Monotone convergence theorem $\implies$

$$\begin{aligned} \theta_\infty &= \lim_{n \rightarrow \infty} P(X_n = 0) \stackrel{\text{MCT}}{=} P(\lim_{n \rightarrow \infty} X_n = 0) \\ &= \text{probability of extinction.} \end{aligned}$$

Continuity of the probability-generating function $\implies$

$$\begin{aligned} G(\theta_\infty) &= G(\lim_{n \rightarrow \infty} \theta_n) \stackrel{\text{CONT}}{=} \lim_{n \rightarrow \infty} G(\theta_n) \\ &= \lim_{n \rightarrow \infty} \theta_{n+1} = \theta_\infty \end{aligned}$$



Supercritical phase $\mu > 1$. Define

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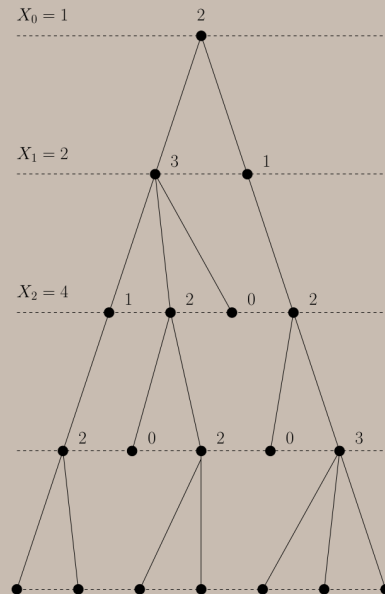
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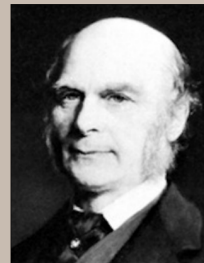
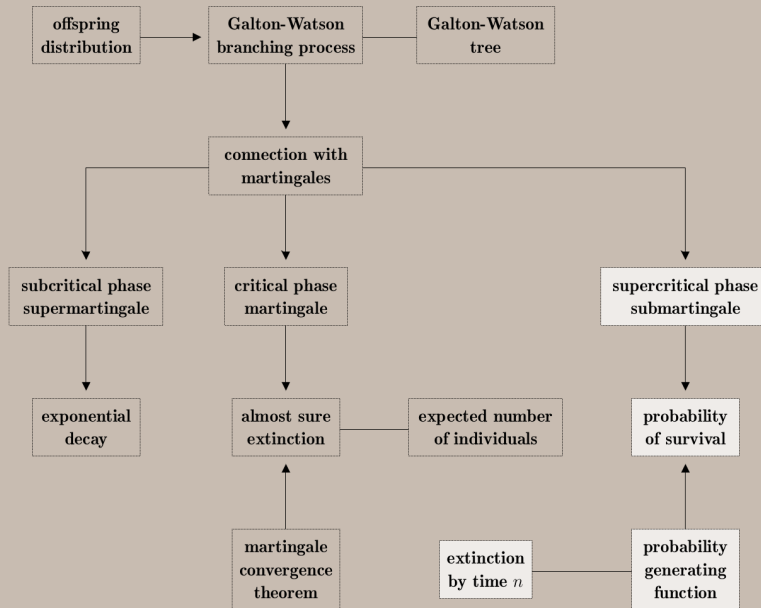
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$\implies$ probability of extinction $\theta_\infty =$ **fixed point** of G .





Francis Galton



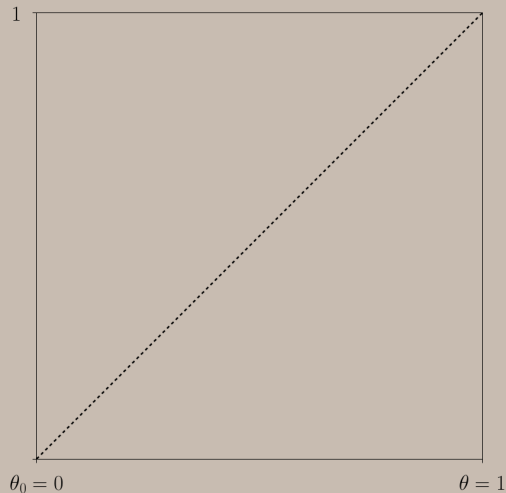
Henry Watson

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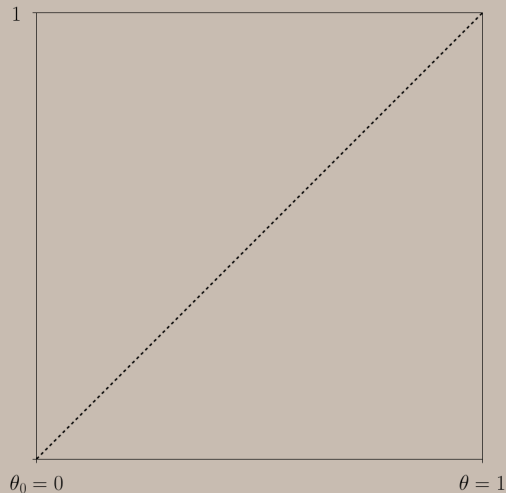
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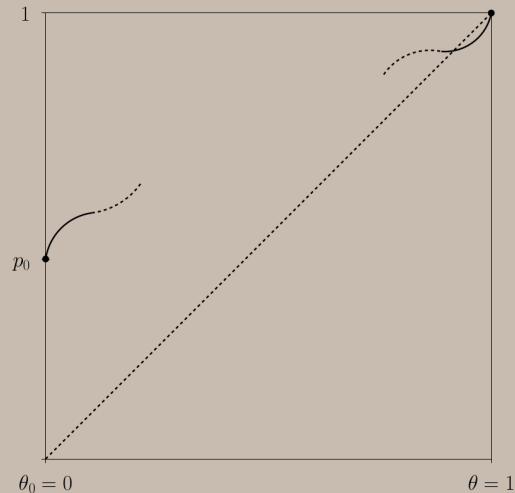
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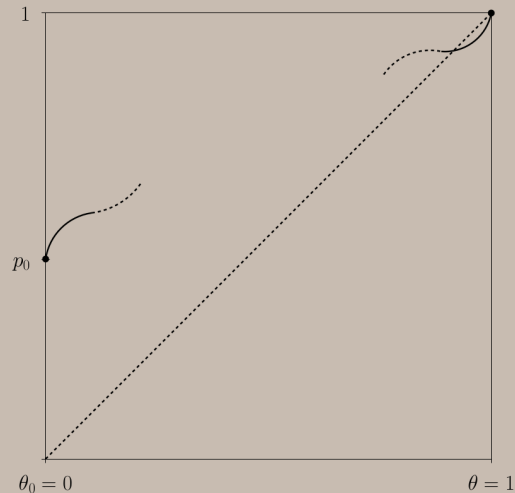
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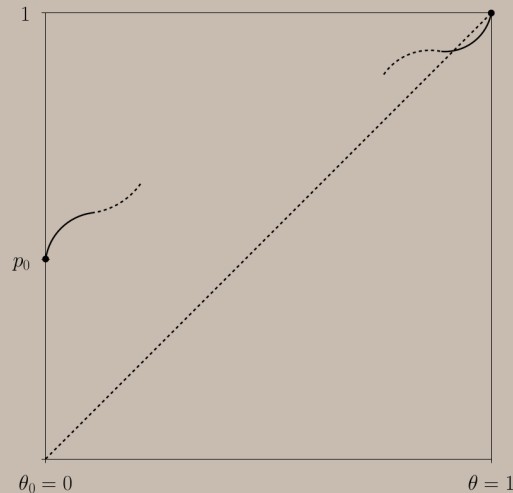
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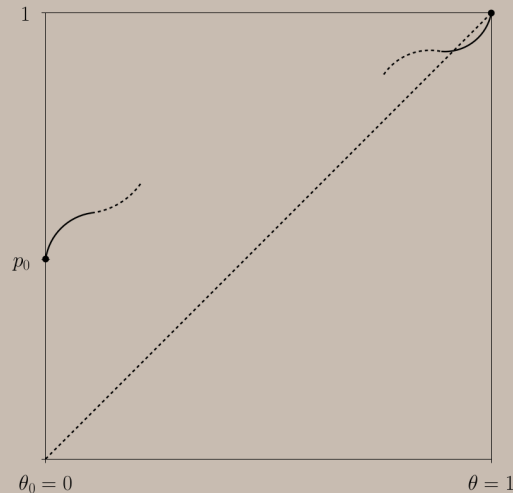
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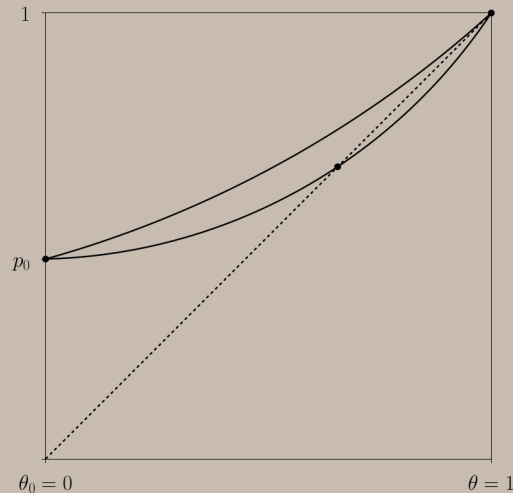
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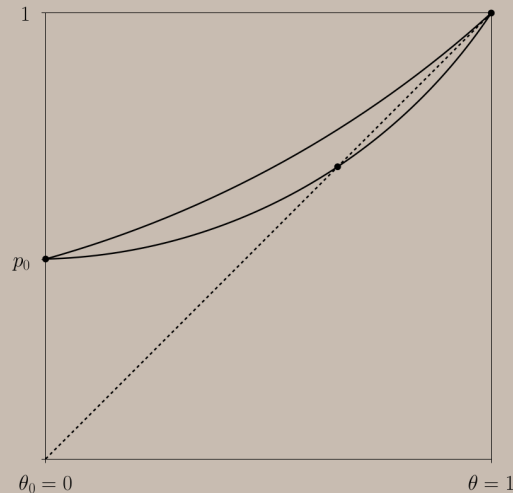
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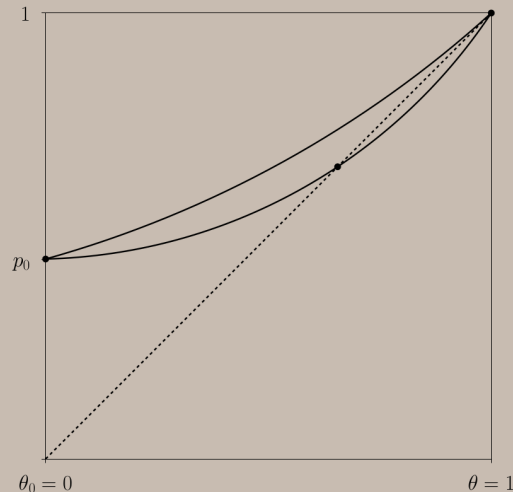
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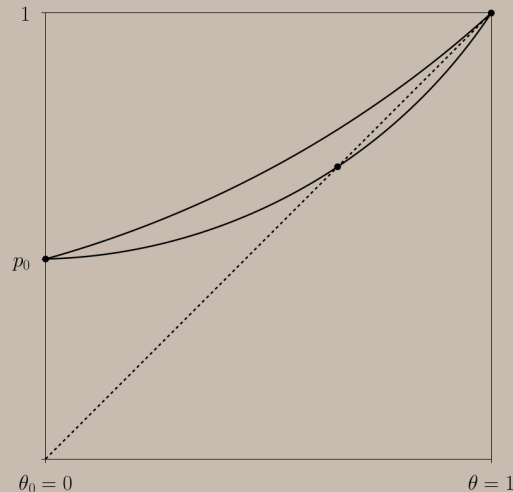
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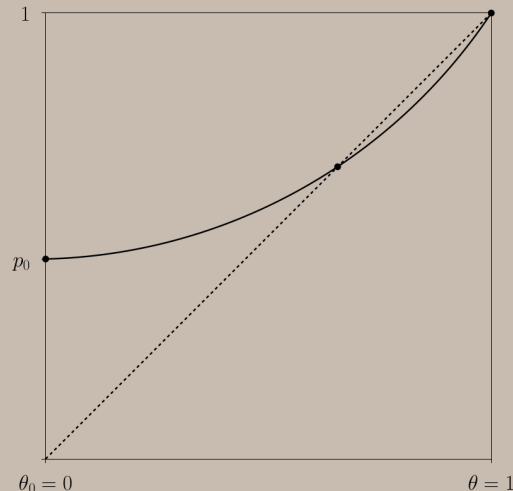
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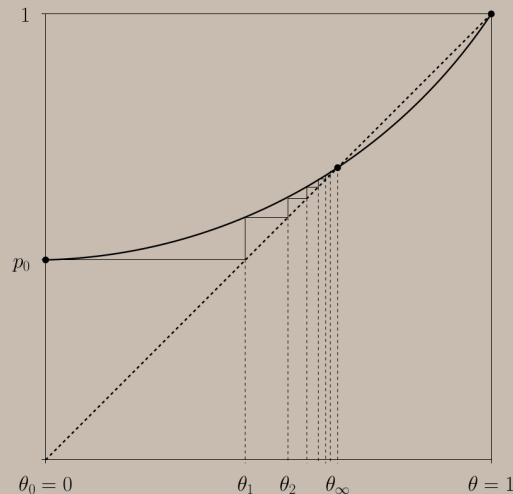
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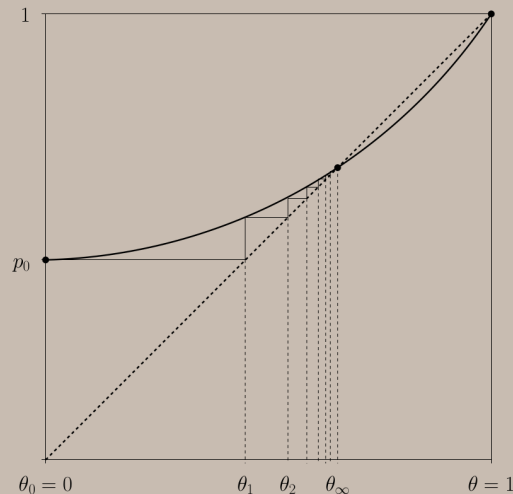
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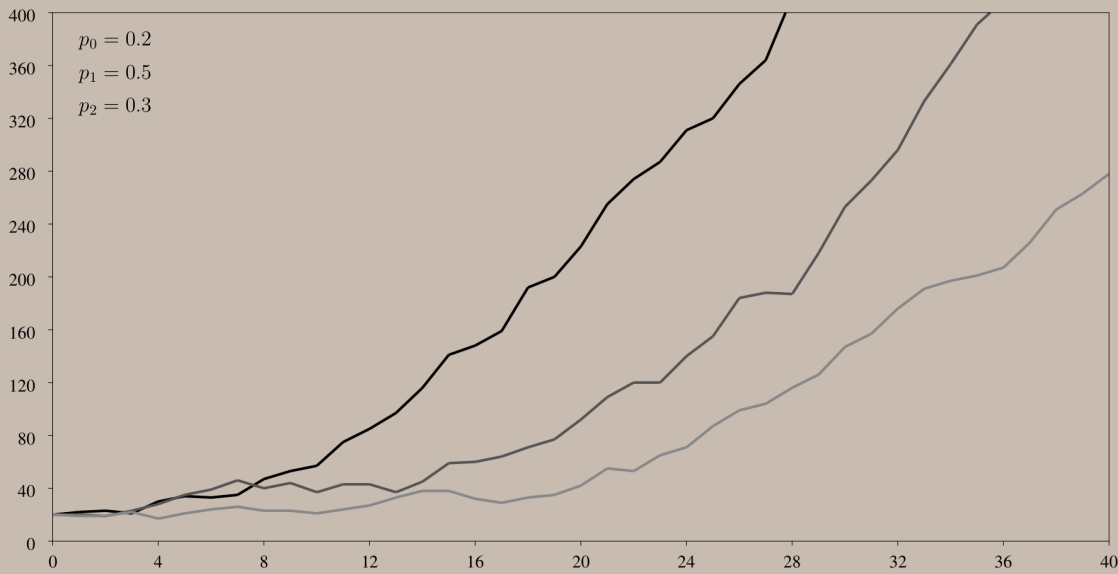
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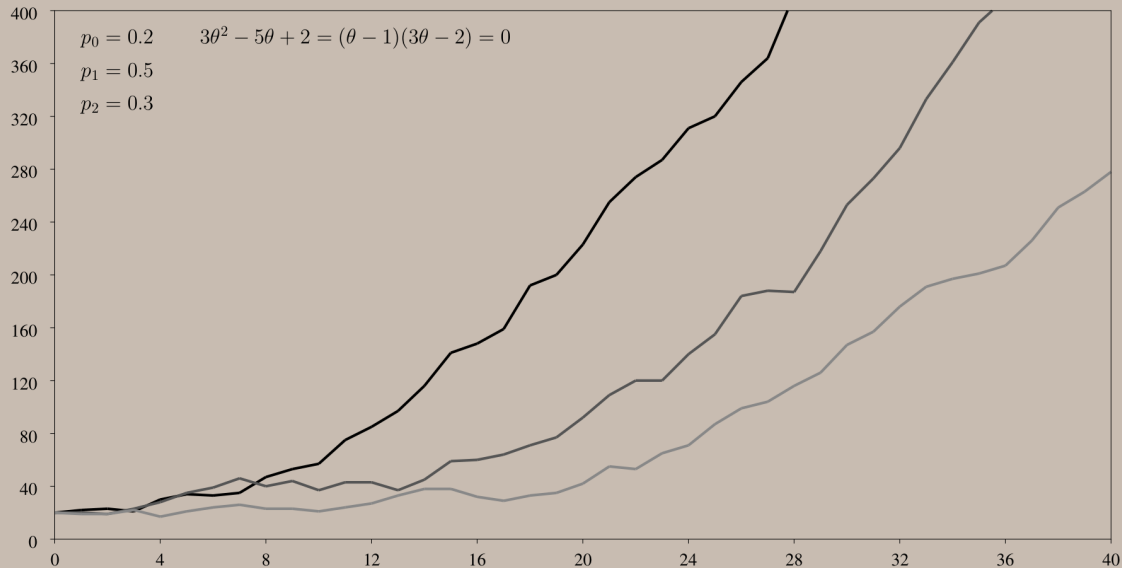
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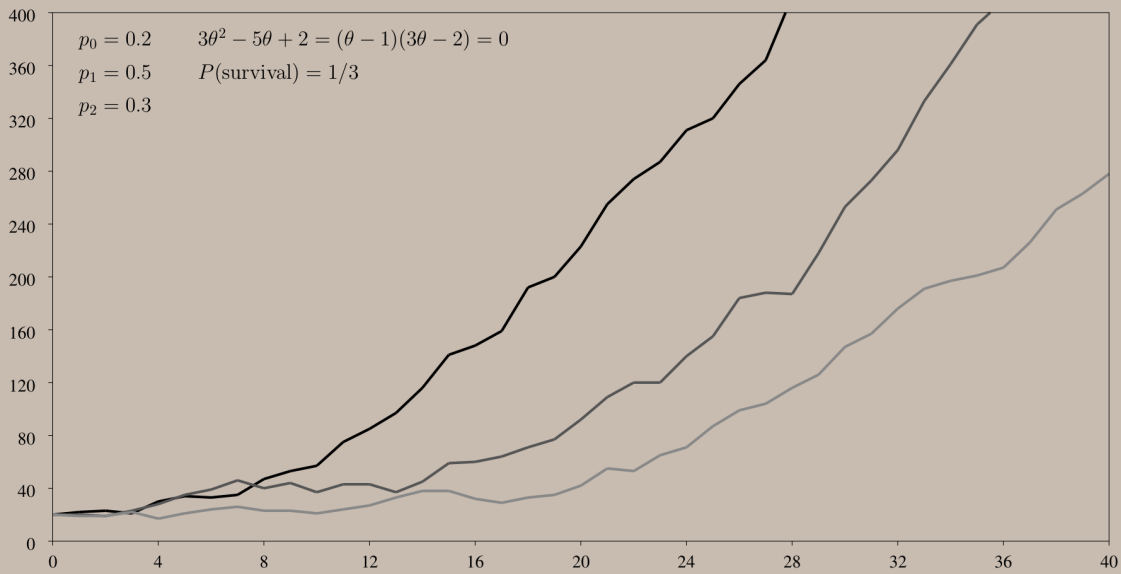
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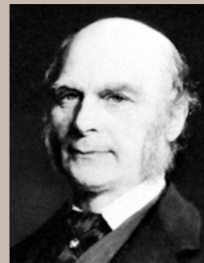
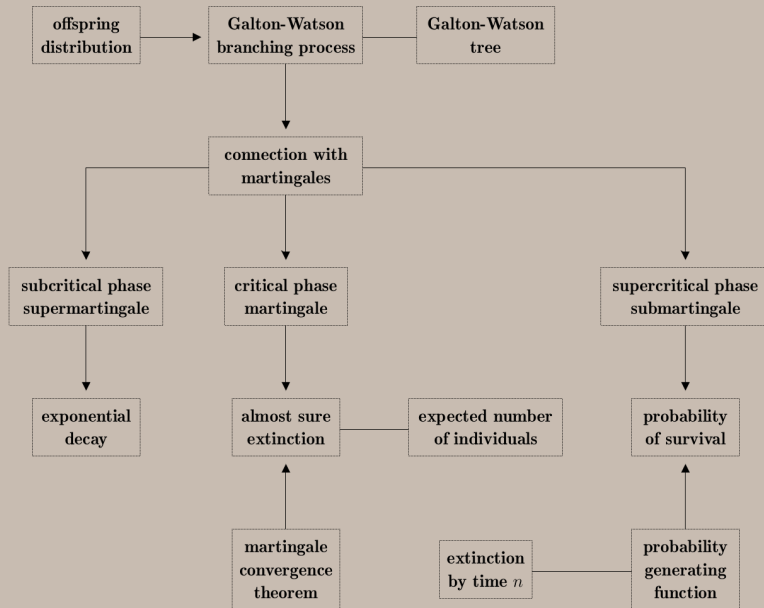
Conclusion: $P(\text{survival}) = 1 - \theta_\infty > 0$.











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